

TULIKA BASU

Prayagraj, Uttar Pradesh

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EDUCATION

VIT Bhopal University

2022 – 2026

B.Tech - Computer Science and Engineering - CGPA - 8.65

Bhopal, Madhya Pradesh

Maharshi Patanjali Vidya Mandir

2019 – 2021

Grade 11-12 - Percentage - 94

Prayagraj, Uttar Pradesh

St. Mary's Convent Inter College

2009 – 2019

Grade 1-10 - Percentage - 96.8

Prayagraj, Uttar Pradesh

PROJECTS

LLM-Driven AgriWeather Alert System | [SpringBoot](#), [Twilio](#), [LLM](#), [MySQL](#), [Scheduler](#)

2025

- Developed an AI-powered system to generate automated, bilingual weather summaries and farming tips.
- Implemented SMS and WhatsApp notifications to provide weather updates for digitally less-literate farmers.
- Leveraged Large Language Model to generate clear and concise, weather reports from raw weather data.
- Enabled multi-language support to serve diverse rural farming communities across regions.
- Integrated voice call functionality, allowing users to access weather updates via speech in Hindi and English.

Blood Bank Management System | [Node](#), [Express](#), [MongoDB](#), [JWT](#)

2024

- Built a backend service to digitalize blood donation management, enabling seamless donor registration, eligibility validation, and appointment scheduling.
- Role-based authentication ensuring controlled access for admins and donors using JWT.
- Preliminary donor screening system to filter out ineligible donors - preventing wasted visits and optimizing center efficiency.
- Personalized donor profiles & admin controls to configure center capacity and scheduling settings.
- Capacity aware slot-booking integrated with QR-based on-site verification to ensure donor authenticity.
- Advanced Donor Filtering and Search by name, blood group, and eligibility status for efficient donor management.

Multimodal RAG System | [Langchain](#), [CLIP Model](#), [Faiss](#), [LLM](#)

2025

- Developed a Retrieval-Augmented Generation (RAG) pipeline for querying information from documents containing both text and images.
- Utilized PyMuPDF for robust document parsing and image extraction.
- Leveraged CLIP Model to create a Unified Vector Store (FAISS) enabling semantic search across multimodal document types.
- Integrated Google Generative AI as a grounded LLM, ensuring contextually accurate responses with source citations, reducing hallucinations.

TECHNICAL SKILLS

Languages: Fluent in Java; familiar with C++, Python; SQL

Coursework: Data Structures & Algorithms, OOPs Concepts, Operating Systems, DBMS

Technologies/Frameworks: Node.js, Express.js, REST APIs, React, MongoDB, FAISS (Vector Search), LangChain, LLM Integration, Prompt Engineering

Developer Tools: GitHub, VS Code, PyMuPDF, Twilio

ACHIEVEMENTS/ CO-CURRICULAR

- Oracle Cloud Infrastructure 2025 Generative AI Professional** (Oracle, 2025) [Certificate]
- Outreach Team **Lead**, BitByBit Club, VIT Bhopal (2024-25)
- Smart India Hackathon (SIH) 2024** college finalist
- Participated in selective internship with the **Madhya Pradesh Police**, gaining hands-on experience in law enforcement operations and cybersecurity awareness (2024)
- Contributor in **Open Source** programs (2024,2025)
- Winner** in Foragers 2.0 (Treasure Hunt) at VIT Bhopal Annual Fest (2023)